

RIBEIRAO PRETO, Brazil

WMO: 836520

Lat: 21.134S

Lon: 47.774W

Elev: 549

StdP: 94.90

Time Zone: -3.00 (SAB)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.5	11.2	3.2	5.1	25.1	5.0	5.8	23.0	7.2	26.9	6.2	25.9	1.9	120	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.8	36.1	21.2	35.0	21.4	34.0	21.7	25.8	31.2	25.1	30.8	24.6	30.1	3.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	20.5	27.7	23.9	20.0	27.4	23.1	19.1	26.8	83.4	31.1	80.3	30.5	77.9	30.3	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	6.0	5.3	DB	6.4	38.4	2.1	1.2	4.9	39.2	3.6	39.9	2.5	40.6	0.9	41.5	(4)
(5)			WB	5.6	26.0	1.8	1.1	4.3	26.8	3.3	27.4	2.3	28.0	0.9	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.2	26.1	26.5	25.9	24.5	21.7	20.7	20.7	22.5	24.6	26.0	25.7	26.1	(6)	
	DBStd	2.99	1.68	1.51	1.61	2.08	2.23	2.10	2.57	2.32	2.63	2.37	1.91	1.57	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	25	0	0	0	1	3	6	11	3	1	0	0	0	(9)	
	CDD10.0	5199	499	462	494	436	362	322	332	387	439	496	470	500	(10)	
	CDD18.3	2181	241	229	236	186	107	78	84	132	190	237	220	242	(11)	
	CDH23.3	20109	1959	2001	1948	1612	907	742	922	1516	2097	2431	1961	2014	(12)	
(13)	CDH26.7	8862	856	917	849	686	293	207	307	676	1094	1238	873	867	(13)	
(14)	Wind	WSAvg	2.4	2.4	2.2	2.1	2.2	2.1	2.3	2.6	2.7	2.8	2.8	2.5	(14)	
(15) Precipitation	PrecAvg	1299	252	228	155	57	54	25	13	16	50	99	150	208	(15)	
	PrecMax	1502	585	433	285	154	115	150	38	87	122	215	300	309	(16)	
	PrecMin	787	79	98	77	0	0	0	0	0	0	23	51	104	(17)	
	PrecStd	190	115	103	52	41	31	42	12	23	40	51	65	55	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	35.9	34.9	34.1	32.1	31.2	32.1	35.0	37.2	38.2	35.2	35.0	(19)	
		MCWB	22.0	23.7	24.2	22.4	20.9	19.6	19.7	19.0	20.2	20.7	23.0	23.3	(20)	
	2%	DB	34.0	34.1	33.8	33.1	30.9	30.1	31.0	33.2	35.9	36.2	34.1	33.9	(21)	
		MCWB	22.5	23.3	23.6	22.2	20.5	19.2	18.8	18.2	19.8	21.1	22.4	22.9	(22)	
	5%	DB	32.9	33.1	32.6	32.0	29.9	29.0	29.9	32.0	34.2	34.9	33.0	32.8	(23)	
		MCWB	22.6	23.2	23.1	22.3	20.1	18.9	18.1	18.0	19.3	20.8	22.1	22.7	(24)	
	10%	DB	31.2	31.9	31.2	30.9	28.2	27.9	28.6	30.8	32.9	33.0	31.8	31.2	(25)	
		MCWB	23.0	23.1	22.9	21.8	19.3	18.5	17.8	17.8	19.1	21.0	22.3	23.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.8	25.9	26.0	25.1	22.7	22.6	21.6	21.4	23.7	26.0	26.8	27.1	(27)	
		MCDB	31.1	32.5	32.0	31.5	29.5	28.1	28.9	31.0	30.8	30.9	30.0	29.8	(28)	
	2%	WB	24.9	25.2	25.1	24.2	21.9	21.2	20.4	20.1	22.5	24.7	25.1	25.7	(29)	
		MCDB	30.4	31.6	30.7	30.2	28.5	26.3	28.2	29.2	30.2	30.5	30.5	30.5	(30)	
	5%	WB	24.3	24.6	24.3	23.5	21.3	20.3	19.5	19.4	21.6	23.5	24.1	24.7	(31)	
		MCDB	29.3	30.6	29.8	29.2	27.1	25.8	27.0	28.1	29.5	29.4	29.7	29.6	(32)	
	10%	WB	23.7	24.0	23.6	22.7	20.4	19.5	18.6	18.6	20.8	22.6	23.2	23.9	(33)	
		MCDB	28.3	29.4	28.9	28.1	25.8	25.3	25.8	27.3	28.0	28.5	28.8	28.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.8	10.6	12.0	12.6	13.7	14.9	15.5	14.3	12.6	11.1	10.3	(35)	
		MCDBR	12.5	12.7	12.6	13.5	14.4	15.7	16.4	17.7	17.2	15.6	13.5	12.5	(36)	
	5% WB	MCWBR	3.8	4.0	4.2	4.6	5.5	6.3	6.1	5.9	5.2	4.3	4.4	4.0	(37)	
		MCWBR	9.9	11.2	11.0	11.7	12.3	13.0	14.5	14.7	13.4	12.2	11.4	10.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.401	0.402	0.393	0.373	0.351	0.353	0.366	0.407	0.506	0.468	0.422	0.410	(40)	
		taud	2.434	2.417	2.449	2.477	2.502	2.494	2.434	2.291	2.013	2.196	2.339	2.393	(41)	
	Ebn at Noon	Ebn at Noon	944	931	914	891	874	851	849	844	797	861	919	936	(42)	
		Edh at Noon	124	124	115	104	94	91	99	123	175	152	135	129	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.63	5.93	5.42	5.07	4.26	4.00	4.27	5.04	5.39	5.81	5.83	5.89	(44)	
	RadStd	RadStd	0.74	0.47	0.48	0.33	0.25	0.28	0.28	0.34	0.51	0.43	0.31	0.37	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page